ANTHONY MOLICA

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EDUCATION

Adrian College, Bachelor of Arts in Data Analytics, Adrian, MI — August 2024 - Expected May 2026

Minor in Computer Science

- Member of the NCHA All-Academic Team

St. Lawrence University, Bachelor of Science in Data Science, Canton, NY — August 2022 - May 2024

- Member of the Academically Elite All-ECAC All-Academic Team - Winter 2023

TECHNICAL SKILLS

Languages: SQL, Python (Pandas, NumPy, Scikit-learn, Matplotlib), R, JSON, Excel
Data Analysis: Exploratory Data Analysis, Feature Engineering, Regression, Classification
Machine Learning: Logistic Regression, Decision Trees, Random Forest, XGBoost
Statistics: Hypothesis Testing, ANOVA, Chi-Square, ROC-AUC, Model Evaluation
Data Handling: Web Scraping, Data Cleaning, Data Visualization

PROJECTS

NHL Shot Data Analytics Capstone — August 2025 – December 2025

- Developed an Expected Goals (XG) Model using XGBoost and Random Forest 2023-2025 short-level data
- Engineered features, identified key predictors, conducted exploratory data analysis, and performed model testing and evaluation
- Identified distance, angle, and individual skill to be the strongest predictors of goal probability, demonstrated that nonlinear models significantly outperform linear models, where 7% of shot attempts become goals

NHL Playoff Performance Prediction on Detroit Red Wings — Spring

- Analyzed 2023-2024 NHL play-by-play data using R to evaluate team discipline, defensive impact, power-play efficiency, and scoring trends through data wrangling, visualization, and comparative league analysis
- Built shot plots and team performance metrics to identify key areas for improvement and translated findings into actionable recommendations for Detroit Red Wings playoff qualification
- Quantified the statistical gaps between Detroit and playoff teams, prioritizing special team efficiency and shot suppression to improve postseason probability

Fitness Tracker Web Application (JavaScript, APIs) — Spring 2026

- Developed a JavaScript-based fitness tracking application with REST API integration to authorize, authenticate, log, and manage user activity data
- Used Postman for API testing and validation, and implemented functionality to track workouts and analyze activity trends over time

Netflix Movie Data Analysis (Python, Machine Learning) — Spring 2026

- Built a machine learning model using Python and Scikit-learn to analyze Netflix movie data and identify patterns in genres, release years, and content features
- Performed data cleaning, exploratory data analysis, and feature analysis to extract insights and visualize trends in streaming content

Work Experience

Professional Hockey Player, Evansville Thunderbolts, Evansville, IN — Spring 2026

- Compete at a professional level while maintaining high-performance physical and analytical preparation
- Utilize video and performance analytics with Instat to evaluate gameplay, opponent tendencies, and team strategy

Interior & Exterior Painting, Distinct Finish LLC, Byron Center, MI — May 2020 – August 2024

- Provided interior/exterior painting services, surface preparation, priming, and precision
- Enhanced customer service skills and developed teamwork in demanding environments